

Introduction to statistical methods for physicists:
a basic crash course

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Chapter 1

Probability and Bayes' theorem

1.1 Probabilities

We need to introduce probabilistic reasoning in science because:

1. many of – if not all – the interesting measurements are affected by various sources of uncertainties;
2. many physical processes must be described as if they present a random component (such processes would be described deterministically if we could keep track of all their degrees of freedom, but we cannot and we must resort to a coarse-grained description).

Case 2 corresponds to, for example, tossing a coin or throwing a dice. Let X denote the outcome of a coin toss. It can result in $X = 1$ or $X = 0$, where by convention we identify 1 with the head and 0 with the tail. *If the coin is fair*, we say that the probability of getting $X = 1$ is

$$P(X = 1) = 0.5$$

and the probability of getting $X = 0$ is

$$P(X = 0) = 0.5 .$$

By completeness, we must have

$$P(X = 0) + P(X = 1) = 1.$$

The generalization to unfair coins is straightforward:

$$P(X = 1) = p , \quad P(X = 0) = q = 1 - p . \quad (\text{unfair coin})$$

The *expected value* of X is

$$\mathbb{E}[X] = 1 \cdot P(X = 1) + 0 \cdot P(X = 0) = p$$

while the *variance* is

$$\text{var}[X] = \mathbb{E}[(X - p)^2] = p(1 - p) .$$

If we toss a coin n times independently, the probability of getting $X = 1$ k times is given by the binomial distribution

$$f(k, n, p) = \binom{n}{k} p^k (1 - p)^{n-k} \quad (1.1)$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is the binomial coefficient. It is a simple exercise to check that the binomial distribution is normalized,

$$\sum_{k=0}^n f(k, n, p) = 1 .$$

The expected value of $\bar{X} = (X_1 + \dots + X_n) / n$

$$\mathbb{E}[\bar{X}] = p$$

while the variance is

$$\text{var}[\bar{X}] = \mathbb{E}[(\bar{X} - p)^2] = \frac{p(1-p)}{n} .$$

For $n \gg 1$, the distribution of $\bar{X}$ tends to a Gaussian with mean $\mu = p$ and standard deviation $\sigma = \sqrt{p(1-p)/n}$,

$$\lim_{n \rightarrow \infty} P(\bar{X}) = \mathcal{N}\left(\bar{X} \left| p, \sqrt{\frac{p(1-p)}{n}} \right.\right) .$$

This can also be expressed by saying that the distribution of $n\bar{X} = X_1 + \dots + X_n$ has expected value np and variance $np(1-p)$, and for $n \gg 1$ tends to $\mathcal{N}(n\bar{X} | np, \sqrt{np(1-p)})$.

The `scipy` library for `python` implements the binomial distribution for $n\bar{X}$ in the `scipy.stats.binom` class. Check the notebook for a comparison between the binomial distribution and the limiting normal distribution.

1.2 Probability densities

The variable $n\bar{X}$ is discrete, which means that it can only take discrete values. On the other hand, a variable X sampled from a Gaussian distribution

$$X \sim \mathcal{N}(X|\mu, \sigma)$$

is continuous, which means that it can take all the continuous values within its support – in the specific case of the Gaussian distribution, the support is the real line. Probability distributions for continuous variables are more conveniently expressed through their probability *density*. For the Gaussian distribution, the probability density is

$$p(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\} .$$

Do not confuse probabilities and probability densities. Probabilities are finite, while probability densities are infinitesimal. For a continuous random variable X , it does not make sense to ask what is the probability that $X = x$. This is always zero, because there are infinitely many outcomes of X ! However, it makes sense to ask what is the probability that the outcome of X lies between x and $x + dx$ for an infinitesimal displacement dx . This is given by

$$P(x \leq X \leq x + dx) = p(x)dx .$$

More generally, the probability that X lies in a finite interval $[a, b]$ is obtained by integrating the probability density,

$$P(a \leq x \leq b) = \int_a^b dx p(x) .$$

The normalization condition is

$$\int_{-\infty}^{+\infty} dx p(x) = 1 .$$

From the consideration of the binomial distribution we learn that a discrete probability distribution can be approximated by a continuous one in the appropriate limit. The last statement is true in a sloppy sense, but cannot be taken rigorously because the binomial expresses *finite probabilities* while the normal is a *probability density*. A more rigorous statement is that the cumulative distribution function of the binomial approaches the cumulative distribution function of a normal distribution in the limit $n \rightarrow \infty$.

1.3 Cumulative distribution

The cumulative distribution is defined as

$$\text{cdf}(x) = P(x_{\min} \leq X \leq x) .$$

For a continuous variable with support on the real line, the definition is equivalent to

$$\text{cdf}(x) = \int_{-\infty}^x dx' p(x') .$$

Cumulative distributions allow us to visually inspect the quantiles of the probability distribution. The quantile p is the point x such that $\text{cdf}(x) = p$. The *median* is the point x that corresponds to the quantile 0.5. The `numpy` library for `python` allows us to compute one-dimensional quantiles of a multidimensional distribution through the `numpy.quantile` routine.

Quantiles are useful statistics because they are invariant under changes of variables. This is easily seen from the definition of the quantile p ,

$$p = \int_{-\infty}^x dx' p(x')$$

and from the fact that, under a change of variable $y = f(x)$, the probability density function transforms as

$$p(y = f(x)) = p(x) \left| \frac{dx}{dy} \right| .$$

From the above equation, you also see that quantiles are invariant only if the transformation $y = f(x)$ is one-to-one.

Note that popular estimators such as the mean or the standard deviations do not share the property of being invariant under reparameterization. The invariance of the quantiles under reparameterization can be easily checked numerically – see the explicit examples in the notebook. Changing variables is most easily performed at the level of the samples. If you draw N samples from a distribution,

$$\{x_1, \dots, x_N\} \sim p(x)$$

then you can draw N corresponding samples from $p(y)$ simply by applying the change of variables to the original samples,

$$\{f(x_1), \dots, f(x_N)\} = \{y_1, \dots, y_N\} \sim p(y) .$$

1.4 Frequentist probabilities

The examples of coin tossing and dice throwing illustrate the so-called frequentist interpretation of probabilities. Roughly speaking, in the frequentist interpretation the probability is given by

$$\text{probability} = \frac{\text{favorable cases}}{\text{possible cases}}$$

in the long run of repeated random processes. For example, a dice has six possible outcomes and, if the the dice is fair, all outcomes are equally probable in the long run; therefore $P = 1/6$ for each $X_i = 1, \dots, 6$.

The frequentist interpretation defines probabilities as the *limiting frequencies* of the possible outcomes, in the limit of infinite repetitions of the experiment. For example, if you throw a fair dice an infinite times, you will get that each outcome is realized $1/6$ times. For the same reason, if you don't know in advance whether the dice is fair and you want to estimate the frequentist probabilities $P(X_1), \dots, P(X_6)$, you must throw the dice as many times as you can and take notes of the frequencies of the outcomes.

Another classic example of frequentist probabilities is the construction of a so-called contingency table. Consider the case of a disease test. From the trial tests before commercialization, it is known that the test produces a ratio p of true positives, i.e., the test was positive only in a fraction p of the diseased patients. This is also expressed through the conditional probability

$$P(\text{positive test}|\text{disease}) = p .$$

At the same time, it is known that the fraction of the true negatives is q ,

$$P(\text{negative test}|\text{no disease}) = q .$$

By completeness, we must also have

$$P(\text{positive test}|\text{no disease}) = 1 - q .$$

and

$$P(\text{negative test}|\text{disease}) = 1 - p .$$

The contingency table is shown in Table 1.1. For concreteness, let us assume

	disease	no disease
positive test	p	$1 - q$
negative test	$1 - p$	q

Table 1.1: Contingency table for a disease test.

that $p = 0.96$ and $q = 0.99$. What is the probability that you have the disease if you test positive?

$$P(\text{disease}|\text{positive test}) = ?$$

We cannot answer this question without knowing the spread of the disease across the population. Let's say that the spread is $d = 0.02$, i.e., 2 % of the population is diseased. Then, by the frequentist interpretation,

$$P(\text{disease}|\text{positive test}) = \frac{\text{favorable cases}}{\text{possible cases}} = \frac{P(\text{disease, positive test})}{P(\text{positive test})}$$

or

$$P(\text{disease}|\text{positive test}) = \frac{pd}{pd + (1 - q)(1 - p)} = 66\%.$$

The above expression is consistent with the two trivial limits

1. when $d = 0$, $P(\text{disease}|\text{positive test}) = 0$ because I already know I am not diseased;
2. when $d = 1$, $P(\text{disease}|\text{positive test}) = 1$ because I already know I am diseased.

As we shall see in the next chapters, a popular alternative interpretation of probabilistic statements is the Bayesian interpretation of probability – also called the subjective interpretation. Bayes' theorem is a key ingredient in the construction of the Bayesian interpretation. However, the theorem can – and is – used also in the context of the frequentist interpretation.

In this chapter, we will illustrate the use of Bayes' theorem in a frequentist setting, since it helps to provide a more pedagogical introduction. Bayes' theorem relates conditional probabilities and conjugate probabilities,

$$P(A, B) = P(A|B)P(B) = P(B|A)P(A) . \quad (1.2)$$

From Bayes' theorem you can prove the following corollary:

- if $\{A_1, \dots, A_n\}$ is a set of complete and mutually exclusive outcomes, then

$$P(B) = \sum_i P(B|A_i)P(A_i) .$$

Using Bayes' theorem and the corollary, it is straightforward to show that

$$\begin{aligned} P(\text{disease}|\text{positive test}) &= \frac{P(\text{positive test}|\text{disease})P(\text{disease})}{P(\text{positive test})} \\ &= \frac{P(\text{positive test}|\text{disease})P(\text{disease})}{P(\text{positive test}|\text{disease})P(\text{disease}) + P(\text{positive test}|\text{no disease})P(\text{no disease})} \\ &= \frac{pd}{pd + (1 - q)(1 - d)} . \end{aligned} \quad (1.3)$$

From the above discussion, it would seem that the frequentist interpretation of probabilities is able to handle external information. However, frequentism can only handle objective statistical information about the population as a whole. If I have subjective reasons, unrelated to the properties of the entire population, to believe that I am diseased with a high degree of probability, there is no place for such information in the frequentist formalism.

1.4.1 The Monty Hall problem

From the corresponding Wikipedia page:

“The problem was originally posed (and solved) in a letter by Steve Selvin to the American Statistician in 1975. It became famous as a question from reader Craig F. Whitaker's letter quoted in Marilyn vos Savant's “Ask Marilyn” column in Parade magazine in 1990:

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

Savant wrote in her first column on the Monty Hall problem that the player should switch. She received thousands of letters from her readers - the vast majority of which, including many from readers with doctorate degrees, disagreed with her answer."

By simulating multiple repetitions of the problem on a computer, you easily find that the probability of winning if you don't switch tends to $1/3$ in the long run, while the probability of winning if you switch tends to $2/3$ - check the notebook. However, the experiments also make it clear that these probabilities are useful to inform your decisions only if you play many times. If you play only once, you better be lucky than rational!

Exercise. Solve the Monty Hall problem using Bayes' theorem.

Chapter 2

Bayesian inference

2.1 The Bayesian interpretation

The Bayesian interpretation of probabilities is that they express degrees of belief in well-defined statements. A Bayesian might legitimately ask questions like “what is the probability that this coin is fair?” – at a more mathematical level, “what is the probability that $p = 0.5$ for this coin?”. For a frequentist these are meaningless questions because the value of p for a particular coin is a fixed – but unknown to us – property of reality, it is not the outcome of a repeated random experiment. In other words, you cannot frame the answer to the question in the form of a long-run frequency ratio

$$\frac{\text{favorable cases}}{\text{possible cases}} .$$

By assigning probabilities to the value of p , a Bayesian does not claim that p physically fluctuates. Bayesian probabilities express the epistemic uncertainty about p . In this sense, Bayesianism is a theory of reasoning that extends the binary logic of True/False statements, also allowing for probabilistic statements.

Bayes’ theorem is also reinterpreted as a logic rule to *update* one’s states of belief. To see how, consider again the case of a disease test. We applied Bayes’ theorem in the form

$$P(\text{disease}|\text{positive test}) = \frac{P(\text{positive test}|\text{disease})P(\text{disease})}{P(\text{positive test})} .$$

According to the Bayesian interpretation:

- $P(\text{disease})$ denotes my belief of being diseased *before* being tested; it is also called the *prior belief* or simply the prior; it is often denoted by the Greek letter π ;
- $P(\text{disease}|\text{positive test})$ denotes my belief of being diseased *after* being tested, as a consequence of the test result; it is also called the *posterior belief* or the posterior;

- $P(\text{positive test}|\text{disease})$ is the probability of getting a given experimental outcome – a positive test – if the argument of the prior and posterior – being diseased – is true; it is also called the *likelihood* because it expresses how likely the experimental result is under the conditional hypothesis; it is often denoted by the symbol $\mathcal{L}$;
- $P(\text{positive test})$ is a normalization factor; it is called *evidence* for reasons that will become clear later; it is often denoted by the symbol $\mathcal{Z}$.

Let us return to the example of a coin toss and apply Bayesian reasoning. If we observe k heads after n tosses, what is the probability distribution of the parameter p ? According to Bayes' theorem

$$p(p|k, n) = \frac{\mathcal{L}(k, n|p)\pi(p)}{\mathcal{Z}(k, n)} . \quad (2.1)$$

Note that we switched to probability densities for $p(p|k, n)$ and $\pi(p)$ because their argument is a continuous variable. Since the evidence does not depend on p , the shape of the posterior density depends only on the likelihood and the prior. The prior must account for any contextual information that I am aware of before taking the measurements. For example, if you know that the coin tosser is a serial cheater, it is justified to place most of the prior density away from the fairness value $p = 0.5$. Vice-versa, if you trust them, it is justified to choose a prior that peaks at $p = 0.5$. This is exactly the kind of information that a frequentist would ban from their theory of inference.

A potential drawback of Bayesian reasoning is that inference quality depends critically on prior choices. For example, if you were so confident that the experimentalist is a cheater and you placed zero prior support at $p = 0$, then the posterior will necessarily vanish at $p = 0$ *even if your prejudices were wrong and the coin was fair!* This argument is often used by frequentists to criticize Bayesian inference. Bayesians acknowledge that this is an intrinsic issue of the theory and that the only solution is to choose the prior distribution as wisely as possible.

2.2 Parameter estimation

For a Bayesian, obtaining the posterior density $p(k, n|p)$ is equivalent to *measuring* p . How is this measurement performed in practice? In a few special cases the likelihood and the prior can be combined in a closed form expression for the posterior. However, in many realistic circumstances, the posterior must be obtained numerically. We give an explicit example in the notebook, where we obtain the posterior numerically by sampling p on a fine grid.

In addition to observing how the posterior changes with the prior, we also observe that, as n increases, the posterior is less and less affected by the prior. This is in agreement with intuition: as n increases, the number of heads k tends to its limiting frequency. Formally, this is explained by the fact that the

likelihood dominates the prior as n increases. This is best seen by taking the log on both sides of Eq. (2.1),

$$\log p(p|k, n) = \log \mathcal{L}(k, n|p) + \log \pi(p) + \dots$$

where the dots indicate constant terms with respect to p . Although the log-prior is insensitive to the data, the log-likelihood has a strong dependence on the quality of the data. At some point, the data become so precise that $\log \pi(p)$ is negligible with respect to $\log \mathcal{L}(k, n|p)$ in most of the relevant parameter space.

Sampling on a grid might be practical when the parameter space is low-dimensional – in the specific case, it is one dimensional! However, as the dimension D of the parameter space grows, the number N of grid points needed to densely cover it scales exponentially with D , $\log N \sim D$. This is the phenomenon known as the *curse of dimensionality*. The development of computers in the second half 20th century made it possible to implement numerical sampling techniques based on a stochastic but efficient exploration of the parameter space. Popular methods include Markov Chain Monte-Carlo and nested sampling. In the notebook we use the `dynesty` library for `python` to sample $p(p)$ with nested sampling.

The use of stochastic samplers is a standard practice nowadays in science, and fostered the spread of Bayesian inference.

2.2.1 Multiple independent datasets

In the previous example, we imagined to perform a single experiment tossing the coin n times. This generates a dataset of the form

$$\mathbf{d} = \{X_1, \dots, X_n\}$$

with $X_1 + \dots + X_n = k$, which we abbreviated with the notation (k, n) . Now, let us consider the case of multiple datasets $\mathbf{d} = \{d_1, \dots, d_M\}$, each independently sampled from the others. We want to sample the posterior p given all observations in each dataset. By Bayes' theorem,

$$p(p|\mathbf{d}) = \frac{\mathcal{L}(\mathbf{d}|p)\pi(p)}{\mathcal{Z}(\mathbf{d})}.$$

The probability of independent events is the product of the individual probability, therefore

$$\mathcal{L}(\mathbf{d}|p) = \prod_{i=1}^M \mathcal{L}(d_i|p)$$

and

$$\log p(p|\mathbf{d}) = \log \pi(p) + \sum_{i=1}^M \log \mathcal{L}(d_i|p) + \dots$$

The posterior can be sampled analogously to the previous case — see the notebook.

2.3 Model selection

In the previous Chapter, we saw that, for large n , (the cumulative of) a binomial distribution is well approximated by (the cumulative of) a normal distribution $\mathcal{N}(k|np, \sqrt{np(p-1)})$. How is the posterior affected if we use the approximate normal likelihood instead of the correct binomial one? If n is sufficiently large, the posterior is not appreciably affected, as expected by the validity domain of the approximation.

However, in real-life cases, we do not know in advance the correct functional form of the likelihood. Therefore, we want to test several models and choose the one that gives a better account of the observed data. This is the subject of model selection. To perform model selection in a Bayesian setting, we formalize the problem and introduce two distinct hypotheses

$$H_0 : \text{binomial likelihood}$$

and

$$H_1 : \text{normal likelihood}$$

Then we rewrite Bayes' theorem to make explicit the dependence of the inference on the assumed hypothesis,

$$p(p|\mathbf{d}, H_i) = \frac{\mathcal{L}(\mathbf{d}|p, H_i)\pi(p|H_i)}{\mathcal{Z}(\mathbf{d}|H_i)} .$$

The key quantity that we need to compute is the odds ratio

$$\frac{p(H_0|\mathbf{d})}{p(H_1|\mathbf{d})} = \frac{\mathcal{Z}(\mathbf{d}|H_0) \pi(H_0)}{\mathcal{Z}(\mathbf{d}|H_1) \pi(H_1)} .$$

The odds ratio naturally factorizes into two terms:

1. the prior ratio $\pi(H_0)/\pi(H_1)$ between the prior beliefs in the two hypotheses;
2. the *Bayes factor* $\mathcal{Z}(\mathbf{d}|H_0)/\mathcal{Z}(\mathbf{d}|H_1)$, which serves to update the prior ratio to obtain the posterior ratio.

Recall that the evidence is given by

$$\mathcal{Z}(\mathbf{d}|H_i) = \int dp \mathcal{L}(\mathbf{d}|p, H_i)\pi(p|H_i) .$$

It expresses the probability of observing the dataset $\mathbf{d}$ if hypothesis H_i is true. In the spirit of Bayesian inference, model selection also depends critically on the assumed priors. Instead, Bayes factors depend only on the data and on the properties of the assumed models, not on the prior beliefs in these models. This is particularly relevant given that model priors depend on multiple factors, including subjective judgment and contextual information available at a given moment in time. Therefore, it is a good practice to report Bayes factors instead

of posterior odds and leave the interpretation open depending on the priors adopted by the reader.

Markov-Chain Monte-Carlo is not optimal to compute evidences, because it is not optimized to sample efficiently from the tails of the posterior distribution. Nested sampling was designed precisely to return an accurate estimate of the evidence together with the posterior samples.

In the notebook we use the `dynesty` sampler to compute the Bayes factor between hypotheses H_0 and H_1 . Specifically, we restrict H_1 to a normal distribution truncated between 0 and n . We use the `scipy` library to define H_0 as the logarithmic posterior mass function `scipy.stats.binom.logpmf`. If we define H_1 as the logarithmic posterior density function `scipy.stats.truncnorm.logpdf` we obtain large Bayes factors in favor of H_1 . Since H_0 is the correct hypothesis, this cannot be correct. The pitfall is that we cannot compute Bayes factors between the model H_0 defined in terms of a posterior mass likelihood (i.e., the likelihood expresses finite probabilities) and the model H_1 defined in terms of a posterior density likelihood (i.e., the likelihood expresses probability densities). For consistency, we must convert the normal likelihood to a posterior mass function. We do so by defining

$$P(k) = P(k - 0.5 \leq x \leq k + 0.5)$$

for H_1 , i.e., we define the posterior mass function as the probability that x is within an interval of unit width centered on k . With this approximation, we recover sensible Bayes factors, although at the expense of the accuracy of the posterior recovered through H_1 . In particular, we get Bayes factors that favor H_0 for small n but approach unity at large n as expected.

Chapter 3

Multivariate Inference

3.1 Correlation reconstruction through a parametric model

Suppose that you conduct an observational campaign of two observables (X, Y) , with the aim of understanding the correlation between X and Y . The exact nature of X and Y is irrelevant to our pedagogical purposes. For example, you can think of the seismic activity (Y) as a function of time (X). We can assume that there is a deterministic physical relation unknown to us because Y is affected by measurement uncertainties: there might be environmental noise from human activities in the vicinity of the seismic detector or the thermal oscillatory motion of the detector components. Schematically

$$s_i = f_{\text{true}}(X_i) + n_i$$

where

1. s_i is the measurement of the displacement from zero at "time" X_i , or the *signal* at X_i ;
2. $Y_i = f_{\text{true}}(X_i)$ is the unknown true underlying value of Y_i and a function of X_i ;
3. n_i is noise contribution to s_i .

In these situations, it is important to model the random processes responsible of noise as accurately as possible to construct the likelihood.

To see how the noise model enters the definition of the likelihood, say that we rescale the variables to arbitrary units, so that noise can be described as a Gaussian noise with variance σ^2 about zero,

$$\log p(n_i) = -\frac{n_i^2}{2\sigma^2} + \dots$$

where dots denote terms independent of n_i . Note that σ sets the relative noise level, more precisely, it sets the root mean squared of the relative noise.

Now, say that you make an educated guess and assume that the physical correlation between Y_i and X_i is described by a linear model

$$Y_i = f(X_i, \alpha, \beta) = \alpha + \beta X_i .$$

Then, *under this assumption*, the likelihood of observing a noise n_i at X_i is

$$\log \mathcal{L}(n_i | \boldsymbol{\theta}) = -\frac{(s_i - f(X_i, \boldsymbol{\theta}))^2}{2\sigma^2} + \dots$$

where we use the compact notation $\boldsymbol{\theta} = (\alpha, \beta)$ to denote the model parameters.

If $d = \{s_1, \dots, s_M\}$ is the dataset comprising the series of observations, the full likelihood is

$$\log \mathcal{L}(d | \boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{i=1}^M (s_i - f(X_i, \boldsymbol{\theta}))^2 + \dots$$

where dots comprise $\boldsymbol{\theta}$ -independent terms.

For definiteness, assume that the true correlation is given by

$$f_{\text{true}}(X) = 0.5 + 1.5X$$

implying that the assumed model reflects the true functional form of the underlying physical law.

Since the model f is linear in X , the problem reduces to ordinary least squares and the likelihood can be written in closed form with respect to $\boldsymbol{\theta}$. However, this is not the case for a general functional model, therefore we proceed as in the general case and use the **dynesty** sampler to sample the posterior $p(\boldsymbol{\theta} | d)$ — see the notebook. We observe that the recovered posterior is not perfectly centered on the true values of the model parameters. This is expected and is due to the random scatter due to noise. We also see that, as expected, the inference becomes more accurate with decreasing σ .

From the posterior samples of $\boldsymbol{\theta}$, we can build the predictive distribution of the model, defined as

$$\{\vec{Y}^a\} = \{f(\vec{X}, \boldsymbol{\theta}^a)\} \quad a = 1, \dots, n_{\text{samples}}$$

and the posterior predictive distribution, defined as

$$\{\vec{Y}^a\} = \{f(\vec{X}, \boldsymbol{\theta}^a) + \text{noise}\} \quad a = 1, \dots, n_{\text{samples}}$$

where the noise is generated from the noise model. They allow us to visualize how well the underlying correlation was recovered and how well simulated data from the posterior resemble the observed data.

By repeating the parameter estimation with a quadratic model,

$$Y_i = f_2(X_i, \alpha, \beta) = \alpha + \beta X_i^2$$

we observe that the estimation converges to well defined posteriors for (α, β) , however, we obtain a poor reconstruction of both the true model and of the data. This is reflected in the large Bayes factor in favor of the linear model.

3.2 Redundant parameters and Occam's razor

Let us now assume a parabolic model

$$Y_i = f_2(X_i, \alpha, \beta, \gamma) = \alpha + \beta X_i + \gamma X_i^2$$

We make the following observations:

1. the posterior distribution shows a correlation between α and γ ;
2. the parabolic model exhibits good model predictive and posterior predictive reconstructions;
3. the Bayes factor shows a moderate preference towards the linear model.

The last observation is commonly referred to as the Bayesian Occam's razor: between similarly good models, Bayes factors tend to prefer the simplest one — in this case, the simplest model is the one with less free parameters.

However, there is a subtle difference between the Bayes factor for the parabolic model and the one for the quadratic model. The quadratic model has the same number of parameters as the linear model, while the parabolic model has one extra parameter. This allows us to artificially inflate (deflate) the Bayes factor by expanding (restricting) the prior range for the extra parameter. We can even make the Bayes factor prefer the parabolic model! This is a common pitfall of model selection in the presence of redundant parameter: unless we have a principled reason to set their prior ranges, the interpretation of Bayes factors is highly ambiguous if not unreliable.

3.2.1 Savage-Dickey ratio

The parabolic model and the linear model are nested, specifically, the linear model is nested in the parabolic one. This means that the linear model can be recovered as a particular instance of the parabolic one — in this case, when $\gamma = 0$. There is an elegant formulation of Bayes factors for nested models, known as the Savage-Dickey ratio. We derive it for the particular case under consideration, but the generalization is straightforward (see for example the proofs at statproofbook.github.io/P/bf-sddr.html and at www.ejwagenmakers.com/2010/WagenmakersEtAlCogPsy2010.pdf.)

Let H_0 be the linear-mode hypothesis and H_1 be the parabolic-model hypothesis. The nesting relation is

$$H_0 = H_1 \cup \{\gamma = 0\} .$$

Then

$$\mathcal{B}_1^0 = \frac{\mathcal{Z}(d|H_0)}{\mathcal{Z}(d|H_1)}$$

But

$$\begin{aligned}
\mathcal{Z}(d|H_0) &= \int d\alpha \, d\beta \, d\gamma \, \pi(\alpha, \beta|H_1) \mathcal{L}(d|\alpha, \beta, \gamma, H_1) \delta(\gamma - 0) \\
&= \int d\alpha \, d\beta \, d\gamma \, \frac{\pi(\alpha, \beta|H_1) \pi(\gamma|H_1)}{\pi(\gamma|H_1)} \mathcal{L}(d|\alpha, \beta, \gamma, H_1) \delta(\gamma - 0) \\
&= \frac{p(\gamma = 0|H_1)}{\pi(\gamma = 0|H_1)} \mathcal{Z}(d|H_1)
\end{aligned}$$

where $p(\gamma = 0|H_1)$ is the marginal posterior density under H_1 with respect to (α, β) . Putting together,

$$B_1^0 = \frac{p(\gamma = 0|H_1)}{\pi(\gamma = 0|H_1)} .$$

This is the Savage-Dickey ratio. Notably, it can be evaluated solely from the prior and the posterior of the larger model, without performing any parameter estimation of the nested model. The expression of the Savage-Dickey ratio makes it explicit that the Bayes factor scales inversely with the prior width of the extra parameters of the larger model.

We evaluate the Savage-Dickey ratio as follows. First, we extract the marginalized samples of γ from the complete set of samples. Marginalizing samples is equivalent to taking only the samples' column corresponding to the parameter(s) of interest. Then we build a density estimator for $p(\gamma|d, H_1)$ using kernel density estimation; specifically, we use a Gaussian kernel. Finally, we use the KDE object to evaluate $P(\gamma = 0|d, H_1)$ and compute the Savage-Dickey ratio.

It is instructive to repeat the parameter estimation under H_1 and the Savage-Dickey calculation, when the prior for γ is bounded by 0 from below. A naive repetition of the above procedure returns an incorrect result. The reason is that KDE does not handle correctly a bounded domain, because by definition it allocates a finite probability density beyond the bounds. A popular solution is to reflect the distribution about the boundary: train the KDE object on an extended set of samples obtained by augmenting the original samples with the corresponding samples reflected about the boundaries. When evaluating the posterior density, multiply by the number of reflections to preserve normalization on the support of the original samples.

3.3 Bayesian coverage

Noise fluctuations are responsible for the bias of the posterior peak with respect to the true values. We can quantify the position of the true values within the marginalized one-dimensional posterior with the corresponding quantiles

$$Q = \text{cdf}(x_{\text{true}}) .$$

$Q = 0$ and $Q = 1$ are the extreme cases when the true value is the left-hand (resp. right-hand) tail of the posterior density. Intuition suggests that, since all

noise realizations are equally likely, there is an equal probability of finding Q within $[0, 1]$,

$$p(Q) = \mathcal{U}(Q|0, 1) .$$

This condition is also known as *Bayesian coverage*. Operatively, Bayesian coverage could be checked by simulating a large set N of datasets and computing Q from each dataset. It can be demonstrated — see, for example, Appendix A.A1 of arxiv.org/abs/2209.06733 — that Bayesian coverage is only guaranteed if the true values of the parameters used to produce each dataset are sampled from the same prior used to sample the posterior distributions. Moreover, Bayesian coverage is a necessary, albeit not sufficient, condition for the sampler to produce consistent posteriors. For this reason, it is commonly used as a diagnostic tool for sampling pipelines. Note that this is based upon the assumption of optimal Bayesian setting, in which you are recovering with the same model used to produce the data and you are properly specifying the noise distribution.

Bayesian coverage can be summarized visually through a P-P plot, also known as quantile-quantile plot, namely the normalized plot of the cumulative distribution of the Q 's. For a continuous variable uniformly distributed within $[0, 1]$ the normalized cumulative distribution is the diagonal line. However, the diagonal is only the expected value of $\text{cdf}(Q)$ in the limit of infinite datasets. For a finite number N of datasets we expect a stochastic deviation from the diagonal.

The expected deviation from the diagonal can be quantified by noticing that, for a fixed quantile level p , Bayesian coverage is equivalent to stating that the true value lies within the p quantile with expectation value p . Therefore, $\text{cdf}(Q)$ follows a binomial distribution. For large N , which is usually the limit in which you perform the P-P plot diagnostic, it is approximated by a normal with mean Q and variance $Q(Q - 1)/N$.

In the notebook we compute the P-P plot by injecting and recovering with the linear model.

3.4 The zero noise realization

A special instance of noise is zero noise realization, where all n_i are zero. Recall that $n_i = 0$ is as probable an outcome of the noise process as any other, therefore, the zero noise realization has the same probability of being realized as any other realization. It is only when we coarse-grain the noise representation that zero noise appears as a “too much ordered” realization, or if you wish a noise realization that appears “unlikely” due to its very low entropy in the coarse-grained phase space.

Exercise. Repeat the parameter estimation with a zero noise realization under the linear and the parabolic models and produce the corresponding corner plots. What do you observe?

Question. It could seem puzzling that, even though the noise realization is zero, the posterior densities have nonzero spreads similar to the ones obtained with a particular noise realization. It follows that the parameter estimation still uses information about the nature of the noise process. Where is the information provided?

Exercise. Show that the zero noise realization corresponds to averaging the log-likelihood over all nonzero noise realizations. This is a simplified version of a similar calculation presented in Section 3.4 of arxiv.org/abs/0904.1017.

Guide to the solution.

1. Write the log-likelihood as

$$\log \mathcal{L}(d|\boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{i=1}^M (s_i - f(X_i, \boldsymbol{\theta}))^2 + \dots$$

where the dots stand for $\boldsymbol{\theta}$ -independent terms.

2. Write $s_i = n_i + f_{\text{true}}(X_i)$ and expand the likelihood in a sum of powers of n_i .
3. Observe that the noise-average of n_i over noise realization obeys

$$\langle n_i \rangle = \int Dn' n'_i = 0 \quad \langle n_i^2 \rangle = \int Dn' n_i'^2 = \sigma^2$$

where Dn' is the measure of n_i .

4. Finally, take the noise-average of the log-likelihood and keep only the $\boldsymbol{\theta}$ -dependent terms.

Chapter 4

Hierarchical Inference